BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

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Abstract

We introduce a new language representation model called BERT, which stands for Bidirectional Encoder Representations from Transformers. Unlike recent language representation models, BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task-specific architecture modifications. BERT is conceptually simple and empirically powerful. It obtains new state-of-the-art results on eleven natural language processing tasks, including pushing the GLUE score to 80.5 percent, MultiNLI accuracy to 86.7 percent, SQuAD v1.1 question answering Test F1 to 93.2 and SQuAD v2.0 Test F1 to 83.1, surpassing human performance on the last two benchmarks. The base model has 110 million parameters and the large model has 340 million parameters.

Keywords: pre-training, bidirectional, transformer, masked language model, fine-tuning

1 Introduction

Language model pre-training has been shown to be effective for improving many natural language processing tasks. These include sentence-level tasks such as natural language inference and paraphrasing, which aim to predict the relationships between sentences by analyzing them holistically, as well as token-level tasks such as named entity recognition and question answering, where models are required to produce fine-grained output at the token level. There are two existing strategies for applying pre-trained language representations to downstream tasks: feature-based and fine-tuning. The feature-based approach, such as ELMo, uses task-specific architectures that include the pre-trained representations as additional features. The fine-tuning approach, such as the Generative Pre-trained Transformer, introduces minimal task-specific parameters and is trained on the downstream tasks by simply fine-tuning all pre-trained parameters.

The two approaches share the same objective function during pre-training, where they use unidirectional language models to learn general language representations. We argue that current techniques restrict the power of the pre-trained representations, especially for the fine-tuning approaches. The major limitation is that standard language models are unidirectional, and this limits the choice of architectures that can be used during pre-training. For example, in OpenAI GPT, the authors use a left-to-right architecture, where every token can only attend to previous tokens in the self-attention layers of the Transformer. Such restrictions are sub-optimal for sentence-level tasks, and could be very harmful when applying fine-tuning based approaches to token-level tasks such as question answering. We therefore propose BERT, which pre-trains a deep bidirectional Transformer encoder using a masked language model objective and a next sentence prediction objective. Figure 1 shows the overall pre-training and fine-tuning procedure.

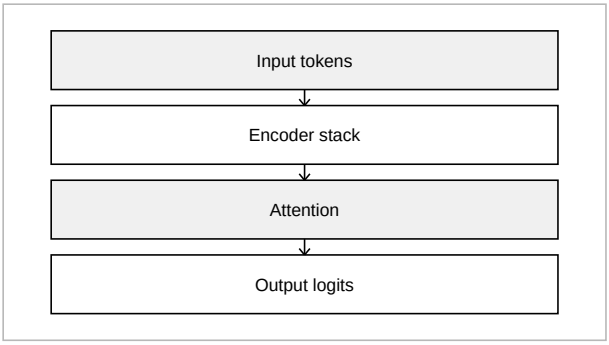


Figure 1: Overall pre-training and fine-tuning procedures for BERT. Apart from output layers, the same architectures are used in both stages.

2 Related Work

There has been a long history of pre-training general language representations, and we briefly review the most widely used approaches in this section.

Unsupervised pre-training methods such as word2vec and GloVe learn word embeddings from unlabeled text, and these embeddings are then used as fixed features in downstream models. More recent approaches pre-train contextual representations. ELMo pre-trains a bidirectional LSTM and uses the concatenation of the left-to-right and right-to-left representations as input features for task-specific models. OpenAI GPT pre-trains a left-to-right Transformer decoder and fine-tunes the entire model on downstream tasks.

Our work differs from these approaches in that BERT is the first model to pre-train deep bidirectional representations. This is achieved by masking a percentage of the input tokens and predicting them from the remaining context, an objective that is sometimes called a cloze task. The model is then fine-tuned end-to-end on downstream tasks with very few task-specific parameters, following the fine-tuning paradigm of GPT but with a bidirectional encoder instead of a unidirectional decoder.

3 BERT

BERT uses a multi-layer bidirectional Transformer encoder based on the original implementation described in Vaswani et al. (2017). Because the use of the Transformer is now commonplace, we will not provide a full background description and refer the reader to the original paper. We denote the number of layers as L, the hidden size as H, and the number of self-attention heads as A. We primarily report results on two sizes: BERT-base (L = 12, H = 768, A = 12, 110 million parameters) and BERT-large (L = 24, H = 1024, A = 16, 340 million parameters). The base model was chosen to have the same model size as OpenAI GPT for comparison purposes, and the large model was chosen to explore the limits of scale.

3.1 Pre-training BERT

Unlike OpenAI GPT, we do not use a left-to-right language model to pre-train BERT. Instead we use two unsupervised tasks, masked language modeling and next sentence prediction, which are both trained on the same corpus. The input representation is capable of representing either a single sentence or a pair of sentences in one token sequence, using special classification and separator tokens. We follow the standard Transformer encoder layout, with layer normalization applied after the residual connection in each sub-layer rather than before it, as described in the appendix.

3.1.1 Task 1: Masked LM

In order to train a deep bidirectional representation, we simply mask some percentage of the input tokens at random, and then predict those masked tokens. We refer to this procedure as a masked language model, though it is also known as a Cloze task in the literature. In all of our experiments we mask 15 percent of all WordPiece tokens in each sequence at random. The final hidden vectors corresponding to the mask tokens are fed into an output softmax over the vocabulary, as in a standard language model. The training loss is the negative log likelihood of the masked tokens, given in Equation (1).

A downside of this procedure is that we are creating a mismatch between pre-training and fine-tuning, because the mask token does not appear during fine-tuning. To mitigate this, we do not always replace the masked words with the actual mask token. The training data generator chooses 15 percent of the token positions at random for prediction. If the i-th token is chosen, we replace it with the mask token 80 percent of the time, with a random token 10 percent of the time, and with the unchanged token 10 percent of the time.

$$L_{MLM} = -\sum_i \log p(mask_i = w_i \mid context) \quad (1)$$

3.1.2 Task 2: Next Sentence Prediction

Many important downstream tasks such as question answering and natural language inference are based on understanding the relationship between two sentences, which is not directly captured by language modeling. In order to train a model that understands sentence relationships, we pre-train for a binarized next sentence prediction task that can be trivially generated from any monolingual corpus. Specifically, when choosing the sentences A and B for each pre-training example, 50 percent of the time B is the actual next sentence that follows A, and 50 percent of the time it is a random sentence from the corpus. The model must predict whether B is the next sentence after A, and the two losses are summed during pre-training.

3.2 Fine-tuning BERT

Fine-tuning is straightforward since the self-attention mechanism in the Transformer allows BERT to model many downstream tasks by swapping in the appropriate inputs and outputs. For each task, we simply plug in the task-specific inputs and outputs into BERT and fine-tune all the parameters end-to-end. At the output, the token representations are fed into an output layer for token-level tasks such as sequence tagging or question answering, and the classification token representation is fed into an output layer for classification such as entailment or

sentiment analysis.

Compared to pre-training, fine-tuning is relatively inexpensive. All of the results in the paper can be replicated in at most 1 hour on a single Cloud TPU, or a few hours on a GPU, starting from the same pre-trained checkpoint. We describe the detailed fine-tuning procedures for each task in the experimental sections.

4 Experiments

In this section, we present BERT fine-tuning results on eleven NLP tasks. We use the GLUE benchmark, SQuAD v1.1 and v2.0, and SWAG as our main evaluation suites. All results are reported on the development sets unless otherwise noted, and test results are obtained from the official evaluation servers.

4.1 GLUE

The General Language Understanding Evaluation benchmark is a collection of diverse natural language understanding tasks. The GLUE results are presented in Table 1. BERT-large obtains an average score of 80.5, which is an absolute improvement of 7.7 percent over the previous state of the art, while BERT-base obtains 78.3. On MultiNLI, BERT-large reaches 86.7 percent accuracy, a 4.6 percent absolute improvement over the previous best result.

Fine-tuning on GLUE is simple: a classification layer is added on top of the classification token, and all parameters are updated for three epochs with a small learning rate. We found that larger batch sizes and longer sequences helped on the smaller GLUE tasks, and that the model is relatively insensitive to the exact learning rate in the range from 1e-5 to 5e-5.

Table 1: GLUE development set results. The average column excludes WNLI, following the benchmark protocol.

System	MNLI	QQP	QNLI	SST-2	Avg
BERT-base	84.6	89.2	91.1	93.5	78.3
BERT-large	86.7	89.3	92.7	94.9	80.5
Prev. best	82.1	88.5	90.1	93.2	72.8

4.2 SQuAD v1.1 and v2.0

The Stanford Question Answering Dataset is a collection of question-answer pairs, where the answer to each question is a segment of text from the corresponding passage. We fine-tune BERT by introducing start and end vectors, and the model predicts the probability of each token being the start or end of the answer span. BERT-large achieves a Test F1 of 93.2 on SQuAD v1.1, which is 1.5 points higher than the previous state of the art, and 83.1 on SQuAD v2.0, which is 5.1 points higher than the previous best result.

SQuAD v2.0 extends the task by allowing for unanswerable questions, which makes the problem more realistic. We treat unanswerable questions by predicting a null answer at the start position, and the model is trained to prefer the null answer when the maximum span score is below a threshold. The large improvement on v2.0 suggests that the bidirectional pre-training is particularly useful for questions that require reasoning over the full passage.

4.3 SWAG

The Situations With Adversarial Generations dataset contains 113k sentence-pair completion examples that evaluate grounded commonsense inference. Given a sentence, the task is to choose the most plausible continuation among four choices. BERT-large reaches 86.3 percent accuracy, an improvement of 27.1 percent over the previous best result. This result demonstrates the value of pre-training for tasks that require commonsense reasoning beyond simple lexical overlap.

5 Ablation Studies

In this section we perform ablation experiments over a number of facets of BERT in order to better understand their relative importance. Table 2 reports the effect of removing the next sentence prediction task, changing the masking strategy, and varying the model size, evaluated on the GLUE development set.

We find that removing next sentence prediction hurts both MNLI and SQuAD, and that a bidirectional model is substantially better than a left-to-right model of the same size. The largest gains come from scale: increasing the number of layers, the hidden size, and the number of attention heads all improve downstream accuracy, though the returns diminish beyond 24 layers. We also experimented with different masking percentages and found 15 percent to be near-optimal for both pre-training loss and downstream performance.

Table 2: Ablation studies on the GLUE development set. Values are average scores.

Configuration	MNLI	SQuAD F1	Avg
BERT-base	84.6	88.5	78.3
No next sentence prediction	83.4	87.3	77.1
Left-to-right only	82.2	85.9	75.4
BERT-large	86.7	90.9	80.5

6 Conclusion

Recent empirical improvements due to transfer learning with language models have demonstrated that rich, unsupervised pre-training is an integral part of many language understanding systems. In

particular, these results enable even low-resource tasks to benefit from deep unidirectional architectures. Our major contribution is further generalizing these findings to deep bidirectional architectures, allowing the same pre-trained model to successfully tackle a broad set of NLP tasks.

We believe that a unified architecture that can be pre-trained on unlabeled text and then fine-tuned for a wide range of tasks is a promising direction for future research. The code and pre-trained models are available for download, and we hope that they will lower the barrier to entry for researchers working on new NLP tasks.

$$L = L_{MLM} + L_{NSP} \quad (2)$$

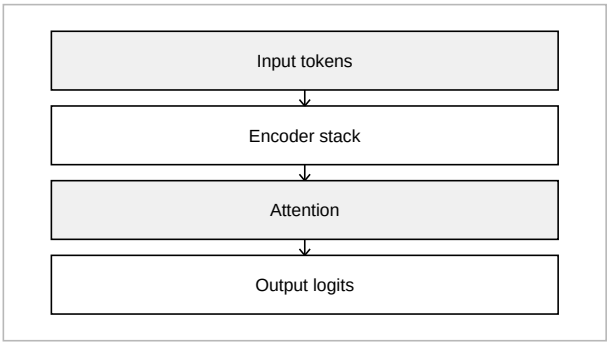


Figure 2: Masked language model training: a subset of input tokens is masked and predicted from the bidirectional context.

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A Appendix: Pre-training Details

We pre-train BERT on the BooksCorpus of 800 million words and English Wikipedia of 2,500 million words. We use a WordPiece vocabulary of 30,000 tokens, a maximum sequence length of 512 tokens, and batches of 256 sequences of length 512, which gives 131,072 tokens per batch. The model was trained for 1,000,000 steps on 64 TPU chips, which took about 4 days. We used the Adam optimizer with a learning rate of 1e-4, beta_1 = 0.9, beta_2 = 0.999, epsilon = 1e-6, and a linear decay schedule with warmup over the first 10,000 steps.

The base model has 12 layers, 768 hidden units and 12 attention heads, for 110 million parameters. The large model doubles the layers and increases the hidden size to 1024 with 16 attention heads, for 340 million parameters. In each Transformer block we apply layer normalization after the residual connection, and we apply dropout with probability 0.1 to all linear projections and the attention probabilities. We use GELU activations and learned positional embeddings, and the classification token is used for sentence-level tasks.